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- **Problem Statement Title- Autonomous Compliance & Regulatory Intelligence System**
- **Team Name- TEAM ACCIO**
- **Team Code- 28012068**
- **Team Member's- Saurabh Gandhi, Vedant Jain, Karan Patil, Ishant Gangwal**



Problem Statement

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Problem Statement - Financial institutions struggle to keep up with rapidly changing regulations from bodies like the Reserve Bank of India, Securities and Exchange Board of India, and Ministry of Corporate Affairs. The current process of tracking, analyzing, and implementing regulatory changes is manual, time-consuming, and error-prone, leading to compliance risks, delays, and heavy financial penalties.

Problems:

1

MANUAL & TIME-CONSUMING PROCESS

Compliance teams spend days reviewing PDFs, comparing versions, and updating policies manually.

2

LACK OF IMPACT VISIBILITY

Organizations struggle to identify which policies, products, or processes are affected by new regulations.

3

HIGH RISK OF ERRORS & DELAYS

Missing even a small change can lead to regulatory penalties, audit failures, and financial loss.



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Proposed Solution

We propose an Autonomous Compliance & Regulatory Intelligence System that automatically monitors regulatory updates, analyzes changes, maps their impact, and generates ready-to-implement policy actions in near real time.

REAL-TIME REGULATORY MONITORING

Continuously tracks updates from Reserve Bank of India, Securities and Exchange Board of India, and Ministry of Corporate Affairs

SEMANTIC CHANGE DETECTION

Identifies meaning-level regulatory changes, not just text differences

POLICY IMPACT MAPPING

Maps changes directly to internal policies, products, and contracts

AUTOMATED AMENDMENT GENERATION

Creates ready-to-use policy updates with proper structure and references

AUDIT-READY REPORTING

Generates complete compliance reports with full traceability

END-TO-END AUTONOMOUS WORKFLOW

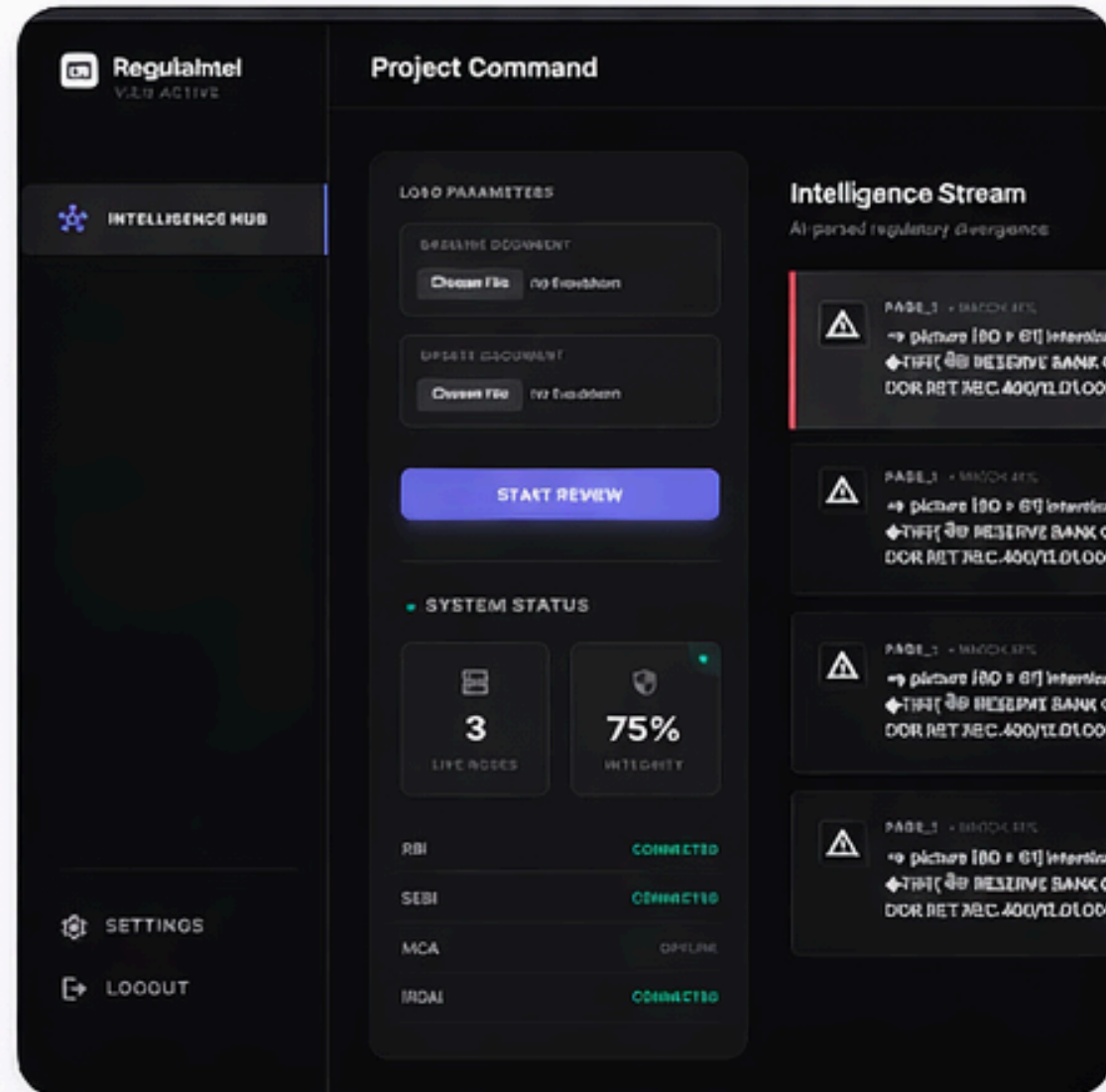
From detection to execution—fully automated with minimal human effort



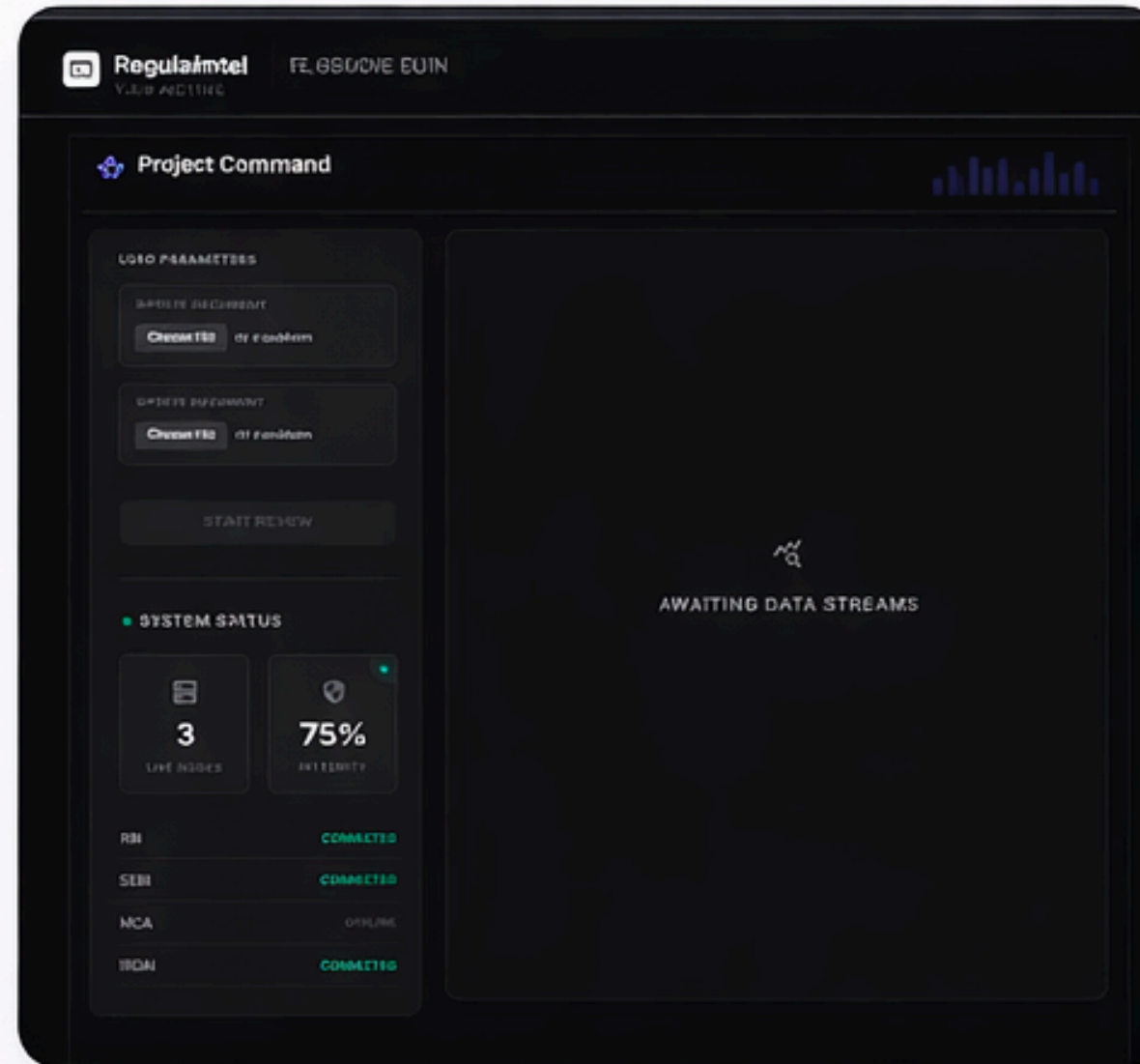
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Prototype

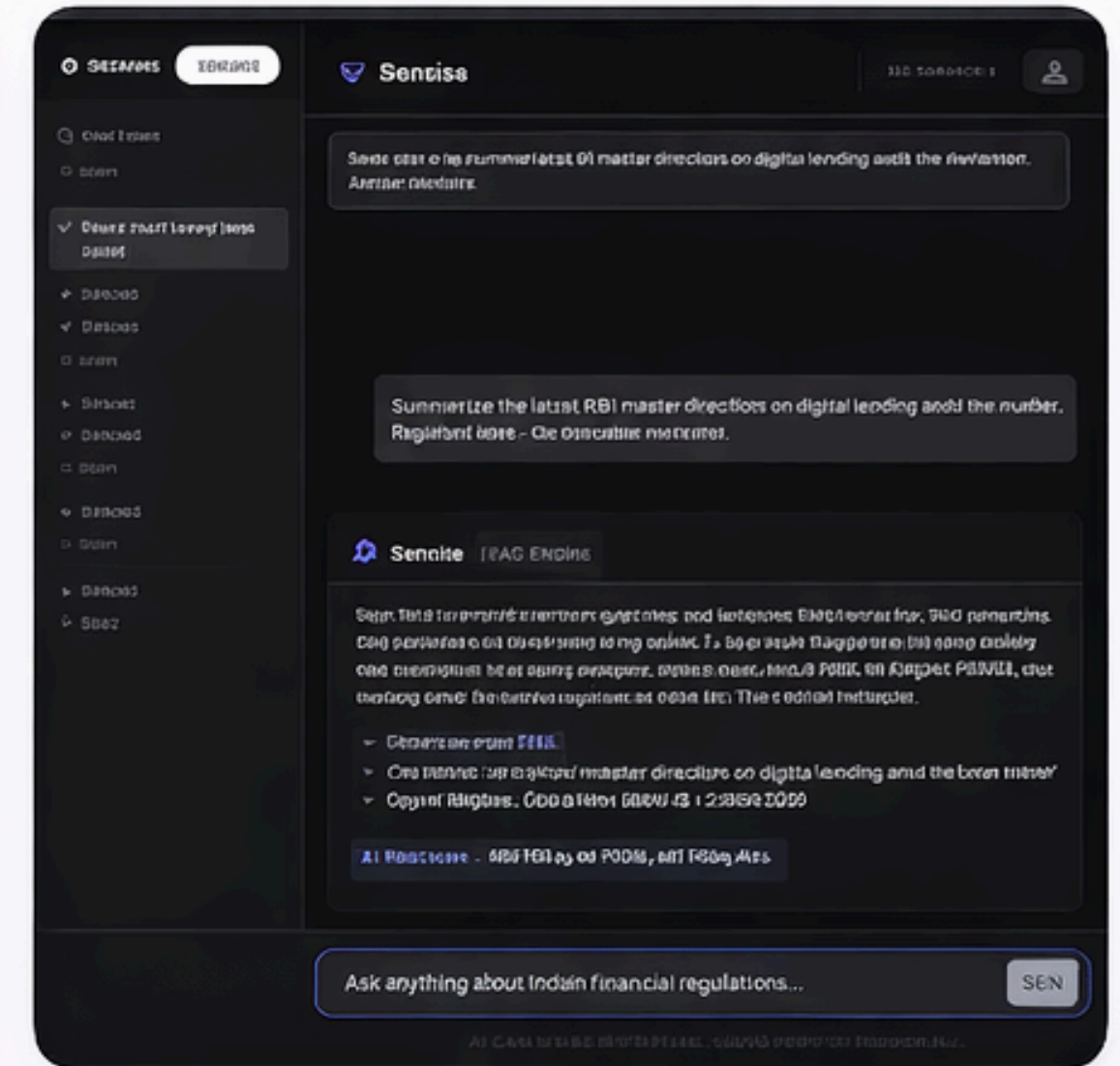
Project Command



Data Loading



RAG Engine





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TECHNICAL APPROACH

Technology Stack



1. Data Ingestion

- Regulatory scraping (RBI, SEBI, MCA, IRDAI)
- RSS feeds + BeautifulSoup
- Incremental update tracking



2. Document Processing

- PDF parsing (PyMuPDF)
- Section-wise chunking
- Text normalization



3. AI & Semantic Engine

- SentenceTransformers (embeddings)
- Hybrid Diff;
 - Cosine Similarity
 - Jaccard Similarity



4. Impact Mapping

- Policy Knowledge Base
- Keyword + Semantic Mapping
- Links changes – policies (KYC, AML, etc.)



5. Backend & Storage

- FastAPI (APIs)
- SQLite + ChromaDB
- Multi-source tracking

SYSTEM ARCHITECTURE



LAYER 1: DATA SOURCES

- RBI / SEBI / MCA / IRDAI circulars
- Regulatory PDFs



LAYER 2: PROCESSING PIPELINE

- Data Extraction
- Semantic Change Detection
- Policy Impact Mapping



LAYER 3: INTELLIGENCE OUTPUT

- Compliance Reports
- Policy Amendments
- Audit Trail



LAYER 4: USER INTERFACE

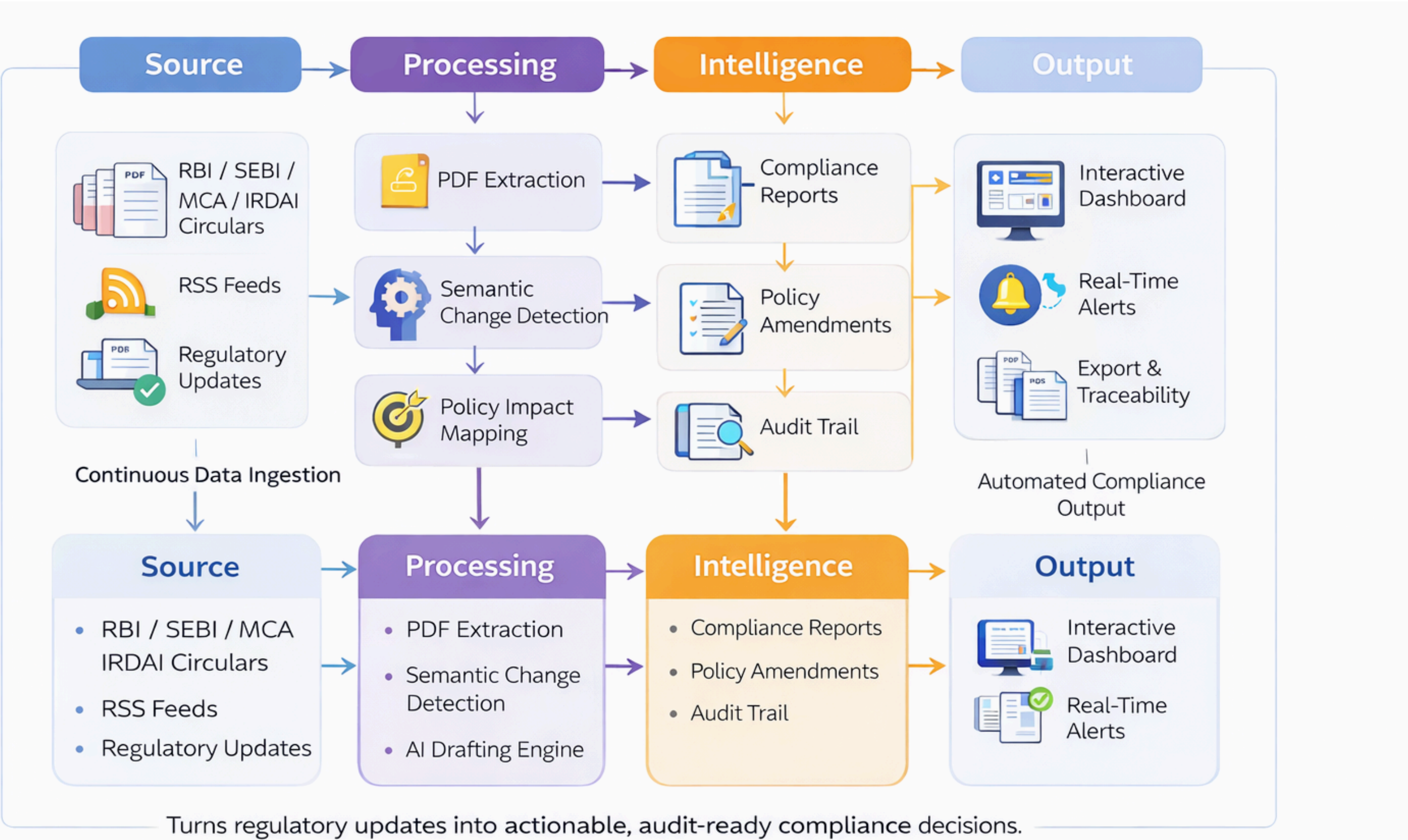
- Interactive Dashboard
- Real-Time Alerts
- Export & Traceability

Autonomous multi-agent pipeline converting regulatory updates into actionable compliance decisions in real time.



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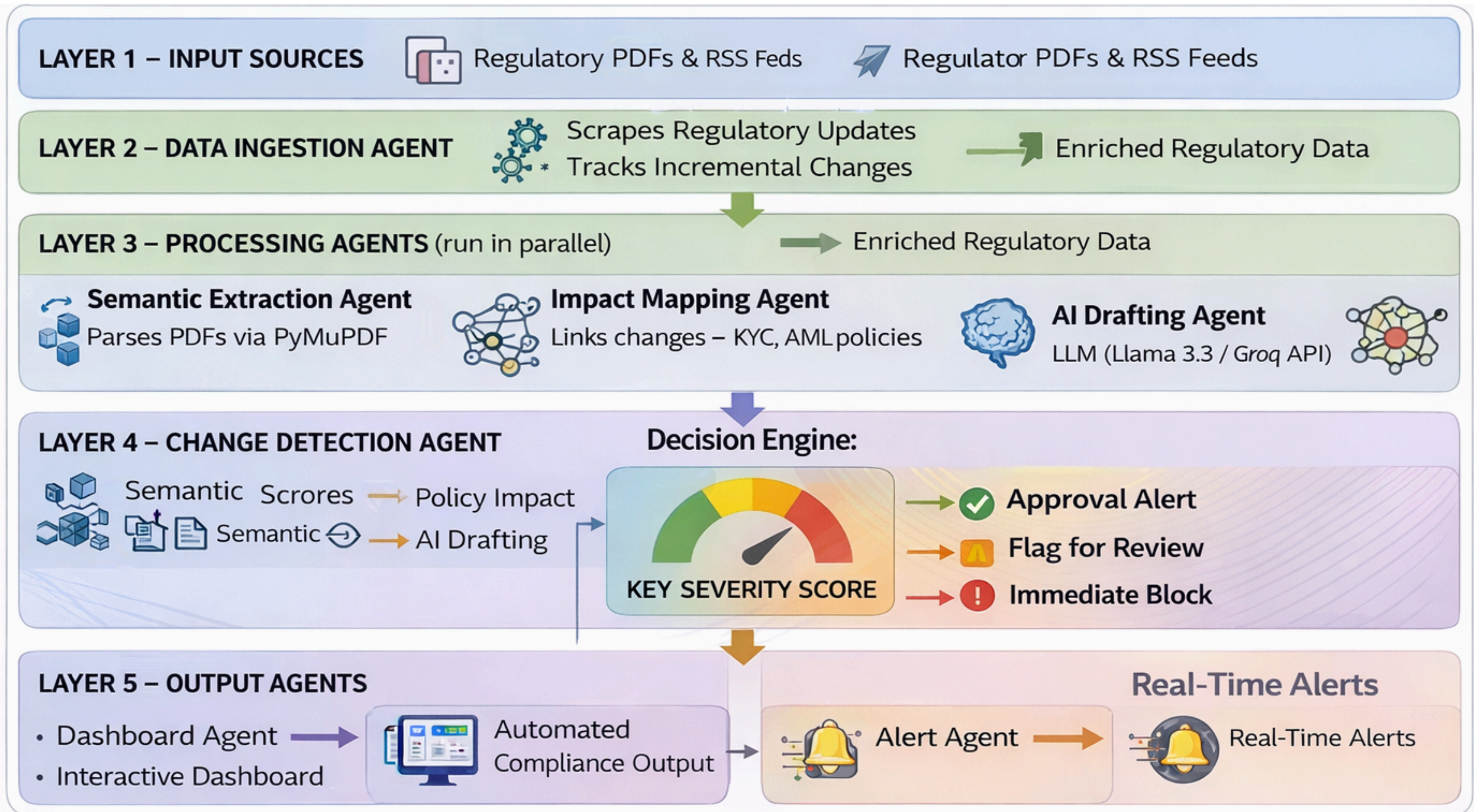
System Architecture





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Agent Architecture





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Key Features

KEY FEATURES

FEATURE 1

Real-Time Regulatory Intelligence

The system continuously monitors updates from bodies like the Reserve Bank of India, Securities and Exchange Board of India, and Ministry of Corporate Affairs, automatically detecting new circulars and eliminating manual tracking, ensuring no regulatory update is missed.

FEATURE 2

Semantic Change Detection Engine

It analyzes regulatory documents at a meaning level using advanced AI, identifying subtle legal changes that traditional text comparison methods fail to capture, thereby reducing the risk of missing critical updates.

FEATURE 3

Intelligent Policy Impact Mapping

The system maps detected changes directly to internal policies, products, and compliance areas such as KYC and AML, providing clear and actionable insights on what needs to be updated.

FEATURE 4

Automated Compliance Drafting

It generates ready-to-use policy amendments with structured formatting and proper references, enabling organizations to quickly implement compliance changes without manual effort.



FEASIBILITY AND VIABILITY

Feasibility:

- **Technical Feasibility:** Uses proven technologies like web scraping, NLP models, vector databases, and LLMs. Your working prototype (PDF parsing, semantic diffing, drafting) already validates core functionality.
- **Operational Feasibility:** Can integrate with existing compliance workflows and dashboards without replacing current systems—acts as a decision-support layer.
- **Scalability:** Modular multi-agent architecture allows scaling across regulators like Reserve Bank of India, Securities and Exchange Board of India, and Ministry of Corporate Affairs, and later global markets.
- **Market Feasibility:** High demand due to increasing regulatory pressure and cost of non-compliance—clear value for banks, fintechs, and enterprises.

Challenges and risks:

- **Unstructured & Inconsistent Data:** Regulatory PDFs vary in format, making extraction difficult.
- **AI Accuracy & Hallucination Risk:** The fraud detection system must handle large volumes of real-time transactions using powerful servers and optimized ML models.
- **Policy Mapping Complexity:** Mapping external regulations to internal policies is not straightforward.
- **Real-Time Processing Load:** Continuous monitoring and analysis can be resource-intensive.

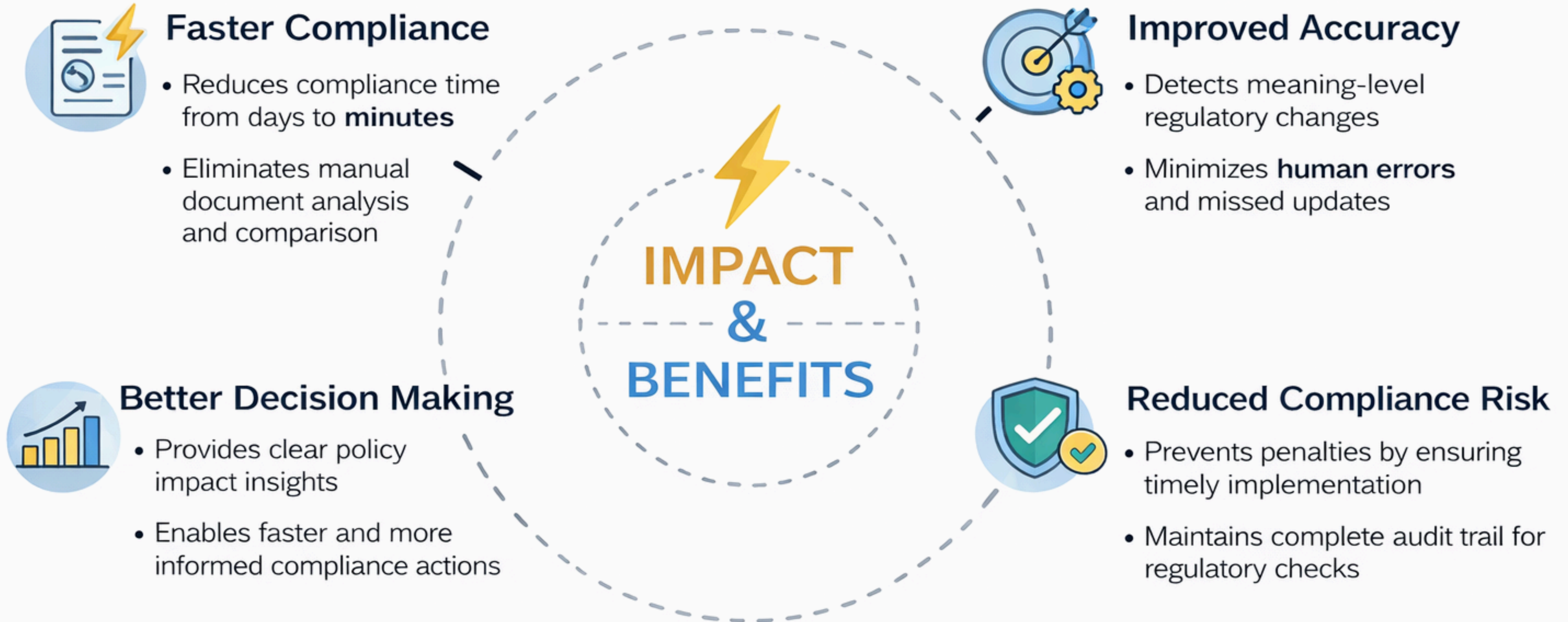
Strategies to overcome the challenges:

- **Robust Data Processing Pipeline:** Use structured parsing, chunking, and validation to handle diverse document formats.
- **Source-Grounded AI + Human-in-the-Loop:** Ensure outputs are traceable with citations and allow final human review for critical actions.
- **Hybrid Mapping Approach:** Combine keyword rules + semantic matching + policy knowledge base for accuracy.
- **Efficient Architecture Design:** Use incremental updates, caching, and optimized pipelines to reduce processing load.



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IMPACT AND BENEFITS



From regulatory updates to **actionable** compliance **decisions in real time**



RESEARCH AND REFERENCES

References

1. Regulatory Sources

- Reserve Bank of India – Regulatory circulars & compliance guidelines.
- Securities and Exchange Board of India – Market regulations & disclosure norms.
- Ministry of Corporate Affairs – Corporate law & compliance updates.
- IRDAI – Insurance regulatory updates.

2. AI & NLP Technologies

- Sentence Transformers (Semantic Similarity Models).
- Large Language Models (Llama 3, GPT-based systems).
- Vector Databases (ChromaDB).
- Natural Language Processing for legal text analysis.

3. Document Processing & Data Handling

- PDF Parsing Tools (PyMuPDF).
- Web Scraping (BeautifulSoup, RSS feeds).
- Data Structuring & Chunking Techniques.

4. Compliance & RegTech Research

- Regulatory Technology (RegTech) frameworks.
- AI in Legal Document Analysis.
- Automated Compliance Systems research papers.

Comparison with Existing Systems

Features	Our Solution	Manual Monitoring	Legacy Compliance Tool
Automatic Regulatory Tracking	✓	✗	✓
Semantic Change Detection	✓	✗	✗
Policy Impact Mapping	✓	✗	✗
Automated Policy Drafting	✓	✗	✓
Audit Trail Transparency	✓	✗	✓



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CONCLUSION

Our **Autonomous Compliance & Regulatory Intelligence System** transforms compliance from a manual, error-prone process into an **intelligent and automated** workflow. By combining **real-time monitoring**, semantic understanding, and **automated** execution, it enables organizations to stay compliant **faster, smarter**, and with greater accuracy.

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Real-Time Monitoring



Smart Decision Making



Reduced Compliance Risk

From regulatory updates → to actionable compliance decisions in real time